

Agriculture: Corn Pest Management Guidelines

Integrated Weed Management

Weeds compete with corn for light, nutrients, and water, especially during the first 3 to 5 weeks following emergence of the crop. It is important to control weeds in a corn field before they are 6 to 8 inches high, which is when they begin to impact corn yields. Late-season weed infestations do not reduce corn yield nearly as much as early weed competition; however, weeds at this time can harbor destructive insect pests such as thrips, which can vector Fusarium ear rot, and armyworms, which can defoliate corn. Weeds also reduce silage feed quality, slow harvesters by causing wheel slippage or clogging, raise grain moisture content, and provide a seed source to infest subsequent crops.

There are over 500,000 acres of corn throughout the state of which 70% is for silage and the rest for grain, with the exception of scattered plantings of sweet corn, ornamental corn, and popcorn. The acreage grown for grain is very price dependent and is usually located in the general proximity of large dairies.

No single weed control regime is effective for all growing conditions. An integrated weed management program utilizes a combination of cultural, mechanical, and chemical methods for consistent, effective weed control. It also helps prevent the development of weed resistance to herbicides and the emergence of a few dominant weeds. A vigorous, competitive crop produced through proper seedbed preparation, variety selection, seeding rates, fertilization, irrigation, cultivation, pest control, and crop rotation is the best defense against weed infestations and competition.

Transgenic Corn. Herbicide-tolerant varieties of corn are being grown in California and provide additional options for weed control. Most transgenic corn being grown is developed by the insertion of genes that transfer tolerance for glyphosate (Roundup), which expands the control options for annual and perennial grass and broadleaf weeds.

Monitoring

To properly manage weeds in corn, survey and identify weeds in the field to be planted to corn while the previous crop is still in the field. Soon after planting, examine the corn field to identify germinated weeds. Make counts as you walk through the field to get a realistic picture of the weed species present, their growth stage, and density. UC ANR Publication 3488, [*Weeds of California and Other Western States*](https://anrcatalog.ucanr.edu/Details.aspx?itemNo=3488), is a good pictorial reference to assist in identifying weeds as is the [weed gallery/\(PMG/weeds intro.html\)](#) of photos available online.

Cultural Control

Cultural practices play an important role in corn weed management. In California a well-managed corn crop is extremely competitive with most weeds. Good cultural practices including timely cultivations often control weeds sufficiently to maximize yields and profit. Keeping fence lines, ditches, and wasteland areas free of weeds will reduce sources of infestation. It is especially important to thoroughly clean tillage equipment and harvesters before entering or leaving a field to prevent spread of weeds from one field to another.

Cultivation is also important in corn weed management because it serves to cut or mechanically bury weeds in the seedling stage before they have a chance to become well established and compete with corn or produce seed. It also serves as a tool for weed resistance management.

Growing corn under no-till or reduced tillage also reduces weeds because the soil is not disturbed, thus reducing the number of seeds that germinate. For weeds that do emerge, postemergent herbicides can be applied.

HERBICIDES

Herbicides are important where weed populations are high, with difficult-to-control weeds, or where it is critical to gain time because cultivation equipment is being used elsewhere. Herbicides reduce the early competition of weed infestation, reduce the seed bank, and reduce the potential for competition in the following crop. Preplant, preemergent, or postemergent herbicides are available that will

selectively control most species of weeds in corn. Select an herbicide based on costs, weeds present,stage of corn growth, soil type, succeeding rotation crop, and adjacent crops.

WEED MANAGEMENT BEFORE PLANTING

Several measures can be taken to reduce weed infestations before the crop is planted, beginning with the selection of a relatively weed-free field. Preparing the seedbed so that it is free of large soil clods provides favorable conditions for corn seed germination and early growth, as well as improved performance of preplant herbicides. The selection of a vigorous growing variety on 30-inch row spacing will help the crop compete with weeds. Uniform plantpopulation densities of 30,000 to 34,000 plants per acre maintain yields andreduce mid- to late-season weed growth by maximizing shading.

Preirrigation or rainfall before planting corn can be useful to germinate weed seeds that can subsequently be controlled by cultivation or postemergent herbicides such as glyphosate, paraquat, or carfentrazone (Shark). Weed sweeps are excellent tools for cultivating johnsongrass, nutsedge, and bermudagrass.

Preplant, preemergent herbicides are applied to the soil surface and mechanically mixed in the soil before the crop is planted. Herbicides applied before the corn emerges offer the advantage of controlling weeds before they compete with the corn when it is in the seedling stage; this is the most critical time in regard to yield reduction. Preplant herbicides such as EPTC (Eradicane), alachlor (Micro-Tech), or metolachlor (Dual Magnum) can be applied broadcast on flat ground and incorporated by discing before beds are formed and the corn is planted, or they can be applied in a band on preformed beds, then incorporated with a rolling cultivator or power tillers.

To maximize the performance of preplant incorporated herbicides, the following are important:

- Preirrigate where feasible.
- Disc in manures before incorporating the herbicide.
- Incorporate to the proper depth: 4 to 6 inches for EPTC and 2 to 3 inches for alachlor or metolachlor. When applying herbicides to beds with a rotary hoe, incorporate two times, once each in opposite directions. Broadcast applications of EPTC require cross-discing for maximum incorporation and effective weed control.
- Time incorporation according to label: EPTC immediately; alachlor and metolachlor, 7 to 14 hours.
- Chop up perennials such as johnsongrass or bermudagrass by cross-discing or mulching before planting.
- Use proper incorporation speed.
- Disturb nutsedge by using sweeps or power mulchers before planting.
- Plant immediately after herbicide application.
- Time the first irrigation to enhance herbicide activity.

WEED MANAGEMENT AFTER PLANTING

After the crop emerges, cultivating with rolling cultivators or sweeps can significantly reduce weeds between the rows, but weeds in the crop row may require further control. Corn plants that are 8 inches or higher have roots that extend well into the furrow. Rolling cultivators cause less root pruning than sweeps or knives, but are less effective on nutsedge, johnsongrass, and bermudagrass. Root pruning can be minimized by staying at least 4 inches from the corn and throwing soil to the plant. (Be careful not to throw hot, sandy soil against tender stalks.)

Weeds not controlled by cultivation can be controlled with a postemergent herbicide application, depending on the weed species present and its growth stage. Postemergent herbicides are most effective when applied to weed seedlings. An over-the-top application can be used but some products or tank mixes require a directed spray on corn larger than 8 to 12 inches in height to keep the herbicide out of the whorl and to minimize the risk of corn injury. Postemergent herbicides commonly used in corn include 2,4-D, bromoxynil (Buctril), carfentrazone (Shark), dicamba (Banvel, Clarity), dicamba/halosulfuron (Yukon), diflufenzopyr (Distinct), halosulfuron (Sandea), metribuzin (Sencor), nicosulfuron (Accent), and foramsulfuron (Option). It is important, however to pay close attention to application guidelines on the labels to avoid phytotoxicity to the crop, especially with carfentrazone (Shark).



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